

OT Sentinel

Demonstration research brief for a low-interaction Modbus, S7 and IEC-104 observatory hosted with a UAE-region deployment model.

SYNTHETIC DATA NOTICE - All quantitative findings in this brief are generated demonstration observations. They validate the pipeline and presentation, but they are not real attacker measurements.

EVENTS	SESSIONS	SOURCES	CONTROL ATTEMPTS
420	418	10	146

Executive summary

OT Sentinel is a low-interaction industrial protocol observatory designed to capture a narrow set of network behaviors without executing attacker instructions or connecting to a real process. The project demonstrates the complete path from protocol telemetry to privacy-reviewed public intelligence.

This release uses a deterministic synthetic dataset so that reviewers can run the dashboard immediately and verify every transformation. Live research will begin only after an isolated cloud deployment is authorized, cost-protected and tested.

The strongest engineering result is not a high attack count. It is the evidence model: a TCP connection is recorded but not labeled as exploitation, while protocol writes, controller commands and program-transfer operations receive explicit ATT&CK hypotheses with confidence and rationale.

Demonstration composition

Events by protocol

modbus		217 / 52%
s7		111 / 26%
iec104		92 / 22%

Leading decoded operations

device_probe		129 / 31%
connection_setup		94 / 22%
write_single		63 / 15%
interrogation		51 / 12%
write_multiple		25 / 6%
single_command		24 / 6%
setpoint_command		17 / 4%
program_download		17 / 4%

Interpretation

The fixture contains 420 events across 418 sessions and 10 pseudonymous sources. 146 events contain synthetic write, command, setpoint or program-transfer operations. These proportions are deliberately varied to exercise dashboard filters; they must not be treated as prevalence estimates.

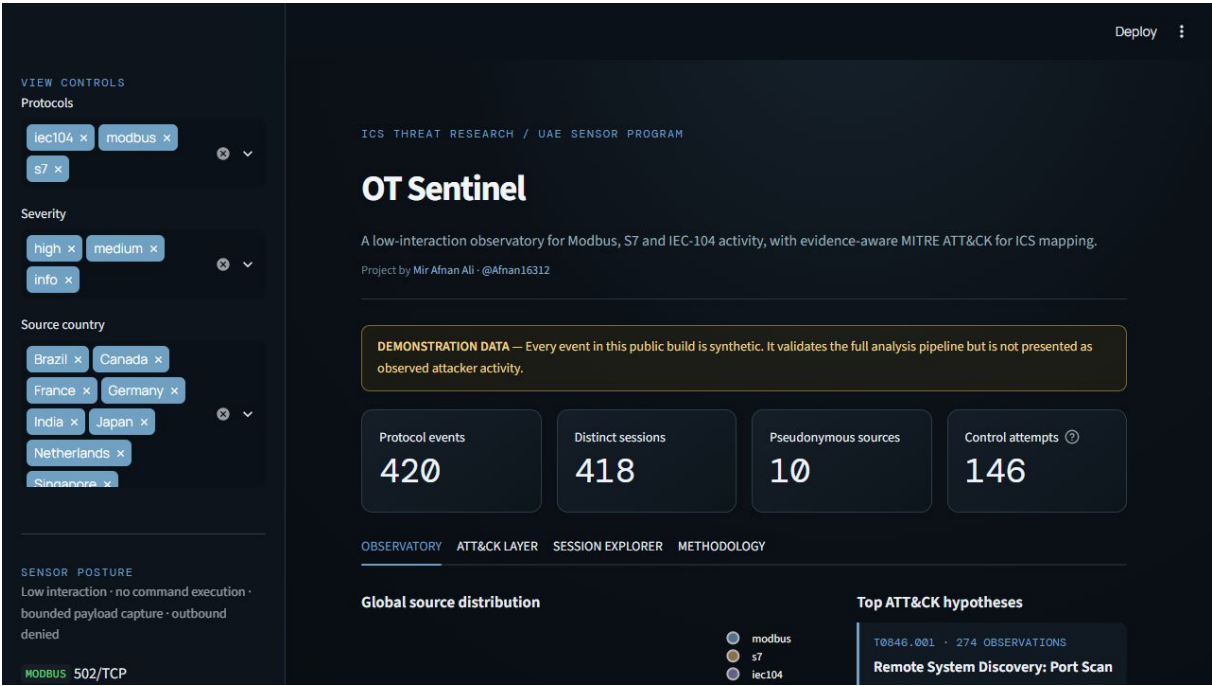
System and analytical design

The Internet-facing surface is intentionally smaller than a conventional SIEM deployment. A single unprivileged Python service reads a bounded payload, performs shallow protocol decoding, emits an inert reply when appropriate and writes an append-only JSON event. It cannot execute a program, invoke a shell or forward an attacker payload.

Collection path

01	COLLECT	Bounded Modbus, S7 and IEC-104 telemetry
02	NORMALIZE	One versioned event schema
03	MAP	Evidence-qualified ATT&CK hypotheses
04	SANITIZE	Salted source IDs; payload removal
05	PUBLISH	Public JSONL, dashboard and report

Public dashboard



The dashboard exposes a global source overview, activity cadence, ATT&CK intensity layer, sanitized session ledger and methodology panel. A permanent banner distinguishes synthetic fixtures from sanitized observations.

ATT&CK evidence layer

Technique counts below are pipeline test outputs, not claims about a threat population. Multiple techniques may map to one event, so counts are not mutually exclusive.

TECHNIQUE	INTERPRETATION	MATCHES
T0846.001	Remote System Discovery: Port Scan	274
T0877	I/O Image	129
T1692.001	Unauthorized Message: Command Message	129
T0836	Modify Parameter	129
T0843	Program Download	17

Confidence controls

A protocol-aware probe may support Remote System Discovery at medium confidence. A Modbus read receives only a low-confidence I/O Image hypothesis because an isolated read does not establish intent. A control write receives high-confidence Unauthorized Command Message and medium-confidence Modify Parameter hypotheses. Exploitation of Remote Services is reserved for a documented signature; it is never inferred from a connection alone.

This distinction improves analytical honesty and makes the output useful for review. Analysts and defenders can inspect not only what was mapped, but why it was mapped and what the evidence cannot prove.

Safety and privacy

The deployment drops Linux capabilities, uses a read-only container filesystem, applies CPU, memory and process limits, and attaches the sensor to an internal Docker network. Raw source IPs and bounded payloads remain private. Publication replaces IPs with salted identifiers, removes payloads and runs an automated forbidden-field check.

Live research plan

The live phase will use one isolated Linux VM in Azure UAE North, funded by Azure for Students or introductory credit. The subscription spending limit will remain active. The experiment will avoid paid Marketplace images, managed SIEM services, Defender upgrades and Log Analytics ingestion.

Collection gates

GATE	REQUIRED EVIDENCE BEFORE PROCEEDING
Cost	Active subscription spending limit and 25/50/75 percent alerts
Isolation	SSH restricted to administrator /32; only ICS decoy ports public
Safety	No outbound route from sensor container; no execution capability
Quality	Known-safe protocol probes create expected events and responses
Privacy	Sanitizer and leakage validator pass; manual review completed
Shutdown	Resource-group deletion tested and collection end date recorded

Publication language

The final report will say: 'Activity observed by an Internet-exposed ICS decoy hosted in Azure UAE North during the stated collection window.' It will not label sources as UAE attackers, claim the observations represent UAE critical infrastructure, or infer physical location from IP geolocation.

Next edition

After a two-to-four-week window, the same report pipeline will replace this demonstration brief with sanitized observed metrics, a stated sample window, data-quality notes, protocol distributions, evidence-qualified ATT&CK results and reproducible limitations. The raw dataset will remain private.

References

MITRE ATT&CK for ICS: <https://attack.mitre.org/matrices/ics/>

ATT&CK STIX data: <https://github.com/mitre-attack/attack-stix-data>

Conpot: <https://github.com/mushorg/conpot>

Azure spending limits: <https://learn.microsoft.com/azure/cost-management-billing/manage/spending-limit>

Streamlit Community Cloud: <https://docs.streamlit.io/deploy/streamlit-community-cloud>